

Decision and Risk Analysis

Decision rule problems

DAEN 427 / ISEN 427 / ISEN 627

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Formulas

Probability model

The prior and data produce the probability to the decision:

$$q = P(H = 1 \mid D).$$

For a next case, q may be posterior predictive.

For a process claim, q may be a posterior tail probability.

Loss model

With zero loss for correct decisions,

$$\mathcal{R}(a_+) = C_{\text{FP}}(1 - q),$$

$$\mathcal{R}(a_-) = C_{\text{FN}}q.$$

Take the positive action when

$$q > \tau, \quad \tau = \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}},$$

$$\tau = \frac{1}{1 + r},$$

where $r = C_{\text{FN}}/C_{\text{FP}}$.

Formulas

Question 1: next case

Decision target:

$$Y_{\text{new}} = 1.$$

Probability:

$$P(Y_{\text{new}} = 1 \mid D) = E(\theta \mid D).$$

Question 2: process state

Decision target:

$$\theta > \theta_0.$$

Probability:

$$P(\theta > \theta_0 \mid D).$$

Q1) Screening a transaction for fraud

An online retailer models the fraud probability θ for transactions in a new risk segment. Before observing segment data,

$$\theta \sim \text{Beta}(2, 18).$$

Among the first 24 transactions, 6 were fraudulent. For the next transaction, the retailer may send it to manual review (a_+) or approve it automatically (a_-).

- a. Find the posterior distribution of θ .
- b. Find the posterior predictive probability that the next transaction is fraudulent.
- c. If $C_{FP} = \$25$ and $C_{FN} = \$180$, find the action threshold and the Bayes action. Verify the decision using expected losses.
- d. Find the critical false-negative cost above which manual review becomes optimal, holding $C_{FP} = \$25$ fixed.

Q2) Intervening in a production process

A food plant tracks the contamination rate θ for a production line. The prior is

$$\theta \sim \text{Beta}(2, 8).$$

Testing finds 8 contaminated samples among 20. Management defines the adverse state as $H = 1$ when $\theta > 0.25$. It may stop and sanitize the line (a_+) or continue production (a_-).

- Find the posterior distribution of θ .
- Compute $q = P(\theta > 0.25 \mid D)$.
- With $C_{\text{FP}} = \$40,000$ and $C_{\text{FN}} = \$120,000$, find the threshold, choose the Bayes action, and compare expected losses.
- If the scientific threshold changes from 0.25 to 0.35, recompute q and determine whether the action changes. Find the critical cost ratio for this revised state.

Note, from R:

```
> pbeta(0.25, shape1 = 10, shape2 = 20, lower.tail = TRUE) → [1] 0.1663049  
> 1 - pbeta(0.35, shape1 = 10, shape2 = 20, lower.tail = TRUE) → [1] 0.4076133
```